

SCH 3U1 - Dilutions of Stock Solutions Worksheet

- Many solutions are prepared in the laboratory from purchased concentrated solutions. What volume of concentrated 17.8 M stock sulphuric acid solution would a laboratory technician need to make 2.00 L of 0.200 M solution by dilution of the original, concentrated stock solution?
- In a study of reaction rates, you need to dilute copper(II) sulfate solution. You take 5.00 mL of 0.050 M $\text{CuSO}_{4(\text{aq})}$ and dilute this to a final volume of 100.0 mL
 - What is the final concentration of the dilute solution?
 - What mass of $\text{CuSO}_{4(\text{s})}$ is present in 10.0 mL of the final dilute solution?
 - Can this final dilute solution of 10 mL be prepared directly using the pure solid? Defend your answer.
- A student tries a reaction and finds that the volume of solution that reacts is too small to be measured precisely. She takes a 10.0 mL volume of the solution by pipet, transfers it into a clean 250 mL volumetric flask containing some pure water, adds enough pure water to increase the volume to 250.0 mL, and mixes the solution thoroughly.
 - Compare the concentration of the dilute solution to that of the original solution.
 - Compare the volume that will react now to the volume that reacted initially.
 - Predict the speed or rate of reaction using the diluted solution compared with the rate using the original solution. Explain your answer.
- A 10.00 mL sample of a test solution is diluted in an environmental laboratory to a final volume of 250.0 mL. The concentration of the diluted solution is found to be 0.274 g/L. What was the concentration of the original test solution?
- As part of a study of reaction rates, you are to prepare two aqueous solutions of cobalt(II) chloride.
 - Calculate the mass of solid cobalt(II) chloride hexahydrate you need to prepare 100.0 mL of a 0.100 mole/L cobalt(II) chloride solution.
 - Calculate how to dilute this solution to make 100.0 mL of 0.0100 mole/L cobalt(II) chloride solution.
 - Write a list of materials, and a procedure for the preparation of the two solutions. Be sure to include all necessary safety precautions.
- In chemical analysis we often dilute stock solution to produce a required standard solution.
 - What volume of a 0.400 M stock solution is required to produce 100.0 mL of a 0.100 mole/L solution.
 - Write a complete procedure for the preparation of this standard solution, including specific quantities and equipment.
- By the additions of water, 30.0 mL of 6.0 M H_2SO_4 is diluted to 150.0 mL. What is the concentration of H_2SO_4 after dilution?
 - 1.2 M
 - 1.5 M
 - 3.0 M
 - 4.8 M
 - 6.0 M
- By the addition of water, 40.0 mL of 8.0 M H_2SO_4 is diluted to 160.0 mL. What is the molarity after dilution?
 - 0.50 M
 - 1.0 M
 - 1.6 M
 - 2.0 M
 - 4.0 M
- A volume, V, of concentrated hydrochloric acid, 18.0 M, is diluted by the addition of 555 mL of water so that the final concentration of acid is 2.40 molar. What is V? Take volumes as being additive.
 - 74 mL
 - 85 mL
 - 133 mL
 - 240 mL
 - 4163 mL

10. By the addition of water, 75.0 mL of 6.0 M H_2SO_4 is diluted to 150.0 mL. What is the concentration of H_2SO_4 after dilution?
- a) 1.2 M b) 1.5 M c) 3.0 M d) 4.8 M e) 6.0 M
11. What volume of 6.00 mol/L nitric acid, $\text{HNO}_3(\text{aq})$, solution is needed to make 4.2 L of 0.15 mol/L HNO_3 solution?
- a) 1.05 L b) 168 mL c) 105 mL d) 214 mL
12. What volume of water must be added to 800 L of 0.130 mol/L solution to dilute it to 0.100 mol/L?
- a) 1840 L b) 1040 L c) 560 L d) 240 L e) 24 L